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# Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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## Abstract

1 While reinforcement learning (RL) has demonstrated remarkable success in en-  
2 hancing the reasoning capabilities of language models, the training dynamics of  
3 RL in LLMs remain unclear. In this work, we quantify and explain the RL train-  
4 ing process through side-by-side empirical measurement and rigorous theoretical  
5 modeling. First, through systematic reasoning-pattern-level and token-level anal-  
6 ysis measured across the entire RL training trajectory, we isolate a striking and  
7 consistent contrast: the intrinsic success rate of each individual reasoning pattern  
8 stays nearly flat throughout training, whereas RL rewrites the probability ranks of  
9 only a sparse subset of critical tokens—fewer than 10% in nearly all conditions  
10 we measure—and it is this small, targeted intervention that reshapes the reasoning-  
11 pattern distribution and drives the measured accuracy gains. This decomposition  
12 makes explicit *which* quantity moves (pattern selection) and which does not (per-  
13 pattern success), across the GSM8K, MATH, and AIME24 datasets and two reward  
14 types. Building on these measurements, we develop a theoretical framework for  
15 the training dynamics of RL under two representative rewards: verifiable reward  
16 (RLVR) and the model’s internal feedback (RLIF). For RLVR, we contrast two  
17 regimes: one in which the model converges rapidly to the optimal reasoning strat-  
18 egy, and one in which convergence time grows super-exponentially, quantifying  
19 how sharply the base model’s reasoning quality separates the two. For RLIF, we  
20 characterize a measurable two-phase pattern—an initial accuracy improvement  
21 (positive with probability 1 in the limit) followed by a 50% chance of converging  
22 to the least accurate reasoning path, i.e. below the base model. Extensive experi-  
23 ments, spanning models from 3B to 32B parameters and multiple mathematical  
24 benchmarks, consistently reproduce these quantitative patterns, advancing both the  
25 theoretical understanding and the practical application of RL for language model  
26 enhancement.

## 27 1 Introduction

28 Recently, state-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-  
29 R1 [14] demonstrate exceptional performance on complex logical tasks including mathematics  
30 [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) serves as a key technique  
31 behind this success, demonstrating the potential to elevate model capabilities to a new level.

32 The success of RL has triggered research into its underlying mechanisms for LLMs. By directly  
33 comparing pass@ $k$  performance between RL-trained and base models, Yue et al. [42] show that  
34 RL post-training struggles to surpass the base model at large  $k$ , suggesting that RL may not elicit  
35 fundamentally new reasoning patterns. From an entropy perspective, Cui et al. [9], Zhang et al.  
36 [45] theoretically prove that RL-based methods can reduce policy entropy, and Cui et al. [9] further  
37 establishes empirically a tight connection between model performance and policy entropy. Comple-

menting these results, Wang et al. [33], Huan et al. [17] report that RL primarily optimizes a sparse subset of critical tokens. Despite this converging evidence, the quantitative dynamics of the RL training process—which quantities move, by how much, and in which direction over training—remain incompletely characterized both empirically and theoretically.

To understand the RL training process, we first present a systematic reasoning-pattern-level and token-level analysis measured across the full RL training trajectory. Specifically, we track two quantities side by side over training: (i) the probability-rank shifts of individual tokens between the base and RL-enhanced models, and (ii) the distribution and per-pattern success rate of reasoning patterns extracted and classified from model responses via LLM-based and rule-based methods. This paired measurement makes the comparison unusually clean. Relative to previous works [17, 42], our experiments provide sharper and more directly quantified evidence: RL leaves the ranks of roughly 90% or more of tokens untouched and rewrites only a sparse critical subset, and this targeted change—rather than any change in the intrinsic success rate of individual patterns, which stays nearly constant—is what *reshapes the reasoning-pattern distribution* and moves overall accuracy. Isolating this contrast between what moves and what stays fixed is precisely what motivates the mathematical framework we develop next.

Grounded in these measurements, we develop a theoretical framework that conceptualizes reasoning as a two-stage question-reason-answer process  $q \rightarrow r \rightarrow a$ : (1) reasoning pattern selection based on the question, i.e.,  $\pi(r|q)$ , and (2) answer generation based on the chosen pattern, i.e.,  $\pi(a|r, q)$ . Using this framework, we analyze the training dynamics of RL in LLMs under two typical reward types: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove convergence to the reasoning pattern  $r^*$  with the highest success rate—exactly the direction our measurements show—and we sharpen this into two quantitatively distinct regimes: a strong base model converges at rate  $\mathcal{O}(1/\epsilon)$ , whereas a weaker base model must first traverse an entanglement stage whose duration grows super-exponentially in a base-model-dependent ratio, a gap of many orders of magnitude between the two regimes. For RLIF, we explain why internal rewards raise accuracy early in training yet, with probability 0.5, ultimately converge to a state *below* the base model, matching the initial-rise-then-collapse trajectory we observe. Additional case studies and simulations reproduce each of these predictions.

The main contributions of this paper are highlighted as follows:

- We conduct systematic reasoning-pattern-level and token-level analysis across the full RL training process. By jointly measuring next-token ranking shifts and LLM/rule-based reasoning-pattern statistics, we deliver a cleaner and more directly quantified account of RL training dynamics than prior work [17, 42]: about 10% or fewer of tokens change rank, per-pattern success rates stay nearly flat, and yet the reasoning-pattern distribution shifts decisively toward high-success patterns—pinpointing pattern selection, not per-pattern optimization, as the driver of measured accuracy gains.
- We develop a formal two-stage mathematical framework that models reasoning as  $q \rightarrow r \rightarrow a$  (question-reason-answer) and analyze the training dynamics of two representative RL rewards. For RLVR, we prove convergence to the highest-success reasoning pattern and separate two regimes with sharply different convergence times— $\mathcal{O}(1/\epsilon)$  for strong base models versus a super-exponential entanglement stage for weaker ones. For RLIF, we prove that accuracy rises at initialization but converges, with a measurable 50% probability, to the least accurate reasoning path, explaining the observed rise-then-degrade trajectory.
- We validate every theoretical prediction through case studies and simulations, and confirm the empirical patterns across models from 3B to 32B parameters and multiple benchmarks. Together this bridges empirical measurement and theory, and offers concrete, quantitatively grounded guidance for RL-based LLM post-training.

## 2 Related Works

**Reinforcement Learning for LLMs** Reinforcement learning has demonstrated remarkable success in enhancing large language models (LLMs), particularly in aligning models with human preferences [25, 48] and in solving complex mathematical and programming tasks [30, 20, 22]. A central component of RL is the reward model, which traditionally relies on human-annotated datasets [26,

25, 2] with extensive training. More recently the paradigm has shifted toward more easily obtainable, verifiable rewards, notably reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]—the two reward types we compare directly in this work. This success has also spurred a family of new RL algorithms, including GRPO [30], DAPO [41], and VAPO [43].

**Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** The success of RL in LLMs has triggered extensive research into theoretical analyses and mechanism interpretation. Initial efforts mainly focused on a high-level analysis of RL, such as how to design better rewards [28, 18, 38] and how to develop novel training algorithms [37, 11, 21]. As RL paradigms shift toward simpler reward structures like RLVR and RLIF, researchers have sought to explain their effectiveness, each isolating one factor: Shao et al. [29] attribute gains to spurious rewards unlocking latent pretraining skills; Agarwal et al. [4] attribute them to entropy minimization; Gandhi et al. [13] to specific cognitive behaviors such as backtracking; and Zhao et al. [46] show that RL fine-tuning amplifies pre-trained behaviors toward a dominant output format, a convergence further analyzed by Wang et al. [34]. Cui et al. [10] through entropy collapse. Our study differs on both axes of evidence: rather than tracking a single scalar indicator, we measure reasoning-pattern selection and per-pattern success side by side over the whole training trajectory, and we pair these measurements with a mathematical framework whose predictions we verify quantitatively.

### 3 Preliminaries

**Reinforcement Learning for LLMs** Let  $\pi_\theta$  be a language model with parameters  $\theta$ , which serves as the policy to be optimized. Given an input question  $\mathbf{x} = (x_0, x_1, \dots, x_n)$ , the policy  $\pi_\theta$  generates an answer  $\mathbf{y} = (y_0, y_1, \dots, y_m)$ . The optimization objective can be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where  $\pi_{\text{ref}}$  is the base reference policy, and  $\beta$  is a hyperparameter that controls the KL divergence to prevent excessive deviation between  $\pi_\theta$  and  $\pi_{\text{ref}}$ .

The reward function  $r_\phi(\mathbf{x}, \mathbf{y})$  can be implemented in various formats, such as a trained reward model [25] or a rule-based scoring function [30]. Throughout this paper we hold this objective fixed and vary only the reward, comparing two representative choices head-to-head: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. This controlled contrast is what lets us attribute the divergent training dynamics we report to the reward signal itself.

In RLVR, the reward function  $r_\phi(\mathbf{x}, \mathbf{y})$  directly evaluates whether the answer  $\mathbf{y}$  matches the correct answer to question  $\mathbf{x}$ . A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Unlike RLVR, RLIF [47, 4] leverages rewards based solely on intrinsic model-derived signals. Following Zhao et al. [47], we consider the RLIF reward as the negative average KL divergence between a uniform distribution  $U$  over the vocabulary  $\mathcal{V}$  and the model’s next-token distribution

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

With both rewards defined under the same objective, the remainder of the paper quantifies and explains how their training dynamics diverge, combining direct empirical measurement with theoretical analysis.

### 4 Experimental Exploration for RL Training

To quantify the RL training process, we conduct systematic experiments and measurement across RL training with different rewards, organized as a top-down comparison at three resolutions. At the coarsest resolution, a training-procedure overview contrasts the two rewards under identical settings: RLVR yields steady, monotone improvement, whereas RLIF rises early and then degrades, sometimes

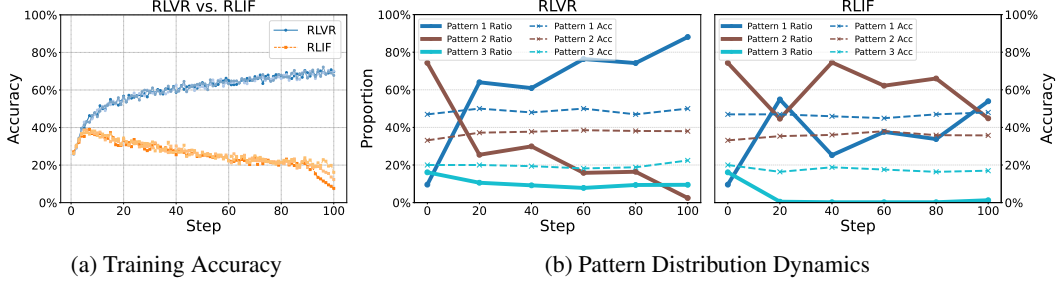


Figure 1: Side-by-side comparison of RLVR and RLIF under identical settings. **(a)** Training-procedure overview on MATH (accuracy vs. training step), with multiple random seeds per paradigm: RLVR delivers stable, monotone gains, whereas RLIF first increases and then decreases. The contrast between the two curves is the headline result. **(b)** Reasoning-pattern-level view, where the solid line is each pattern’s share of all responses (left axis) and the dotted line is that pattern’s success rate (right axis). Under RLVR the distribution shifts steadily toward the higher-accuracy patterns and away from the lower-accuracy ones, tracking the accuracy gains in (a); under RLIF the distribution never stabilizes. Patterns are sorted and named from highest to lowest success rate, so a rightward shift in mass corresponds directly to a drop in expected accuracy.

ending below the untrained base model. Zooming in, a reasoning-pattern-level analysis explains the gap—RLVR steadily concentrates probability on the highest-success patterns, while RLIF fails to specialize. At the finest resolution, a token-level analysis pins down the mechanism: the pattern shift is driven by rank changes in only about 10% or fewer of the tokens. Read together, these three views give a cleaner and more directly quantified picture of RL training dynamics than prior work [17, 42], showing that RL optimizes only a sparse subset of critical tokens and thereby *reshapes the reasoning-pattern distribution* to move model performance.

#### 4.1 Training Procedure Overview for RLVR and RLIF

We first analyze the complete training procedures for both RLVR and RLIF. To make the comparison strictly controlled, we fix the base model to Qwen2.5-3B [39] and the training set to MATH [16], holding every other setting identical across the two algorithms so that the reward is the only variable; we repeat each run under multiple random seeds to confirm the trajectories are consistent rather than seed-specific.

The resulting trajectories separate cleanly, as shown in Figure 1a. RLVR traces a stable, monotone curve: accuracy on MATH improves continuously and gradually converges. RLIF follows a qualitatively opposite arc—an initial climb in the early steps followed by a pronounced decline, in some runs falling below the original base model it started from. The consistency of this rise-then-degrade signature across seeds, contrasted against RLVR’s steady gains, is what motivates the finer-grained analysis that follows.

#### 4.2 Reasoning-pattern Level Analysis

To explain the performance gap measured between RLVR and RLIF, we decompose accuracy into two factors at the reasoning-pattern level: how often each pattern is selected, and how often each pattern succeeds. To define the patterns, we collect responses from the base model and use GPT-4o [19] to group them into distinct categories by keywords and logical structure, with a description per pattern. We then track both factors—the pattern distribution and the per-pattern success rate—at fixed checkpoints across the RL training procedure, so that any change in overall accuracy can be attributed to one factor or the other.

Specifically, we track the dynamics of reasoning-pattern selection and its accuracy for both RLVR and RLIF. Figure 1b reports the Number Theory tasks from MATH [16], with matching results across additional models and datasets in Appendix A. Reading the selection factor against the success-rate factor yields three consistent findings:

- Under RLVR, selection moves monotonically toward the highest-success patterns; because per-pattern success is fixed (third finding), this reallocation alone accounts for RLVR’s steadily rising overall accuracy.
- Under RLIF, the selection distribution never settles and does not concentrate on the more effective patterns, mirroring the unstable accuracy trajectory in Figure 1a.
- The per-pattern success rate stays essentially flat throughout training for *both* methods, isolating pattern selection—not per-pattern optimization—as the sole moving factor behind the accuracy difference.

Taken together, the two rewards differ entirely in the selection factor while sharing the same fixed success-rate factor, so the RLVR–RLIF performance gap is fully attributable to their distinct pattern-selection dynamics. We next probe, at the token level, what mechanism drives these selection changes.

### 4.3 Token-level Analysis

To identify the mechanism behind the pattern-selection dynamics above, we follow Huan et al. [17], Yue et al. [42] and measure change at the token level. We sample base-model responses for questions from GSM8K [6], MATH [16], and AIME [7], then, holding these responses fixed, compare each token’s rank—the position of its probability among all vocabulary tokens given the preceding context—between the base and RL-enhanced models. The comparison is stark (Table 1): across all three datasets, both rewards, and every checkpoint from step 20 to step 100, the fraction of rank-shifted tokens stays around or below 10% (RLVR 5.1–7.3%, RLIF 5.6–10.1%), so roughly 90% or more of tokens are left in place in every case. RL thus rewrites only a sparse set of critical tokens while the bulk of the reasoning process is untouched—exactly the surgical intervention that can move pattern selection without altering per-pattern success.

Across all three resolutions, the measurements converge on a single, quantitatively grounded conclusion: *RL optimizes only a sparse subset of critical tokens (under 10%), thereby reshaping the reasoning-pattern distribution to move model performance, while the intrinsic success rate of individual patterns stays nearly constant throughout training.*

Beyond deepening our understanding of RL training dynamics, this clean separation between what moves and what stays fixed is precisely the structure our mathematical framework is built to explain.

Table 1: Fraction of tokens whose probability rank changes after RL, by dataset, reward, and training step. Values stay below 10% in nearly every cell; RLIF consistently perturbs slightly more tokens than RLVR at matched steps, and the fraction grows only gradually with training.

| Task   | Method | Step20 | Step40 | Step60 | Step80 | Step100 |
|--------|--------|--------|--------|--------|--------|---------|
| GSM8K  | RLVR   | 5.2%   | 6.0%   | 6.6%   | 7.1%   | 7.3%    |
|        | RLIF   | 6.8%   | 8.0%   | 8.9%   | 9.5%   | 10.1%   |
| MATH   | RLVR   | 5.3%   | 5.8%   | 6.1%   | 6.5%   | 6.6%    |
|        | RLIF   | 6.1%   | 7.0%   | 7.7%   | 8.3%   | 8.8%    |
| AIME24 | RLVR   | 5.1%   | 5.7%   | 5.9%   | 6.1%   | 6.3%    |
|        | RLIF   | 5.6%   | 6.5%   | 7.4%   | 8.2%   | 8.6%    |

## 5 Theoretical Characterization of RL Training Dynamics

Building directly on the measurements above, we now derive a theory that reproduces them. The empirical decomposition—selection moves, per-pattern success does not—maps onto a two-step reasoning process: the model first selects a reasoning pattern and then deduces an answer, with RL acting almost entirely on the first step. Within this framework we recover each observed effect: for RLVR we prove convergence to the highest-success pattern and separate two regimes whose convergence times differ by orders of magnitude, matching the strong-vs-weak base-model contrast; for RLIF we explain both the early accuracy rise and the subsequent collapse below the base model, matching the measured rise-then-degrade trajectory.

### 5.1 A Theoretical Framework for Reasoning Models

**Abstract Reasoning Process.** We formalize the reasoning process as follows: given a question  $q$ , the model (1) selects a reasoning pattern  $r$  from candidate patterns  $\mathcal{R} = \{r_1, r_2, \dots\}$  and (2) generates a final answer  $a \in \mathcal{A} = \{a_1, a_2, \dots\}$  accordingly. The model selects reasoning pattern  $r_i$

with probability  $p(\mathbf{r}_i|q)$ , and each reasoning pattern has a distinct success rate. Within this framework, we can reformulate the RLIF optimization objective (Eq. 3.3) as follows<sup>1</sup>:

$$r_\phi(\{\mathbf{r}, \mathbf{a}\}, q) := - \left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(\mathbf{a}_j|q, \mathbf{r})) \right) \quad (5.1)$$

**Policy Parameterization.** Let  $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$  denote the vocabulary set and  $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$  represent the set of output sequences. Given a question  $q$ , the language model produces a distribution over output sequences  $\mathbf{y} \in \mathcal{Y}$  autoregressively:

$$(\text{general policy}) \quad \pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{l=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \prod_{l=1}^{|\mathbf{y}|} \text{softmax}(f_\theta(\mathbf{x}, \mathbf{y}_{<l}))_{\mathbf{y}_l}, \quad (5.2)$$

where  $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$  is a function parameterized by  $\theta$ , and the model predicts the  $l$ -th token based on the previous context  $\mathbf{y}_{<l}$ . The next token follows the distribution  $\text{softmax}(\mathbf{z})_v := \exp(\mathbf{z}_v) / \sum_{v' \in \mathcal{V}} \exp(\mathbf{z}_{v'})$  for  $\mathbf{z} = f_\theta(\mathbf{x}, \mathbf{y}_{<l})$ .

Due to the complexity of practical models, establishing optimization guarantees for understanding RL training dynamics has proven very challenging [3, 23]. Following previous works [27, 24, 10, 45], we consider a *tabular policy* parameterization, which can be viewed as a special case of Eq. 5.2 where each output is assigned its own trainable logit for the corresponding last token, i.e., for  $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ :

$$(\text{tabular policy}) \quad \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \pi_\theta(\mathbf{y}_l|\mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

where  $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$  is the column of  $\theta$  corresponding to  $\mathbf{y}_{l-1}$ .

**Optimization for RL.** For our analysis of training dynamics, we consider the tabular policy from Eq. 5.3 with the optimization objective  $\phi_{\text{RL}}(\theta)$  in Eq. 3.1. We analyze the policy gradient in the small learning rate limit using gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where  $\theta(t)$  represents the parameters of the policy  $\pi_{\theta(t)}$  at training time  $t$ , initialized with  $\theta(0) = \theta_{\text{ref}}$ .

Our measurements showed that the success rate of each individual reasoning pattern stays nearly flat during training—the one factor that does not move. We attribute this to the model’s architectural constraints: reshaping the question-to-pattern mapping is far easier than reshaping the pattern-to-answer mapping. We encode this measured invariance directly as an assumption:

**Assumption 5.1.** *The success rate for each reasoning pattern  $\mathbf{r}_i \in \mathcal{R}$  to provide the correct answer  $\mathbf{r}^*$ , defined as  $\pi_\theta(\mathbf{a}^*|q, \mathbf{r}_i)$ , remains constant during training.*

In the remaining part of this section, we adopt Assumption 5.1 and use  $p^*(\mathbf{r}) = \pi_\theta(\mathbf{a}^*|q, \mathbf{r})$  to denote the fixed success rate for the given pattern  $\mathbf{r}$ . We first derive the optimal policy for the RL objective (Eq. 3.1) under the general autoregressive policy (Eq. 5.2). We then analyze RLVR and RLIF training dynamics using the tabular policy (Eq. 5.3).

## 5.2 The Optimal Policy for RL

Our measurements showed RLVR steadily reallocating selection toward higher-success patterns while RLIF fails to specialize; we now show the two optimal policies explain exactly this contrast:

**Proposition 5.2.** *Suppose we maximize the RL objective (Eq. 3.1) using a general autoregressive policy (Eq. 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|q) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where  $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q)$  is the normalizing coefficient, and  $R(\mathbf{r})$  denotes the reasoning path reward. Specifically:

<sup>1</sup>For simplicity, we omit the normalizing coefficient  $1/|\mathcal{Y}|$  from Eq. 3.3, which can be treated as a constant  $1/2$  in our framework.

- For RLVR,  $R(\mathbf{r})$  equals the success rate of reasoning pattern  $\mathbf{r}$ , i.e.,  $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$ .
- for RLIF,  $R(\mathbf{r})$  equals to the confidence for the reasoning pattern  $\mathbf{r}$  for final answer distribution, i.e.,  $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$ .

Proposition 5.2 makes the RLVR–RLIF contrast precise at the level of the optimal policy. Assuming  $\pi_{\text{opt}}$  lies within the general autoregressive policy space (which always holds for both LLMs and the tabular policies discussed later), RLVR raises the probability of pattern  $\mathbf{r}$  exactly when its success rate  $p^*(\mathbf{r})$  is high enough that  $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$ —so its reward is tied directly to correctness. RLIF, by contrast, rewards only the confidence of the final-answer distribution and cannot distinguish the correct answer  $\mathbf{a}^*$  from incorrect alternatives. This single structural difference explains the measured divergence: RLVR’s optimum consistently improves accuracy, whereas RLIF’s optimum carries no such guarantee and can settle below the base model.

Moreover, since  $\beta$  is typically small (e.g., 0.001) in practice [26, 5, 44], as  $\beta \rightarrow 0$ , we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the policy converges to a deterministic strategy that always selects the reasoning pattern  $\mathbf{r}$  with highest  $R(\mathbf{r})$ , regardless of how the initialized reference model chooses reasoning patterns. However, due to the non-convexity of the optimization landscape, the dynamics of how RLVR reliably finds high-reward patterns and why RLIF may exhibit early performance improvements remain unclear. To address this, we next analyze the training dynamics of both methods under a tabular policy (Eq. 5.3), providing further insight into the RL training process.

### 5.3 Training Dynamic Analysis for RLVR

Our measurements show that the size of RLVR’s improvement depends heavily on the base model’s capacity; the theory below turns this into a sharp, quantified dichotomy. We track the probability of the *optimal reasoning pattern*—the pattern  $\mathbf{r}^*$  with the highest success rate for reaching the correct answer  $\mathbf{a}^*$ ,  $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$ —and find two distinct regimes under gradient flow. In Regime 1,  $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$  rises steadily to 1. In Regime 2, the model first passes through an entanglement stage in which a suboptimal pattern  $\mathbf{r}' \neq \mathbf{r}^*$  stalls progress toward  $\mathbf{r}^*$ , and only afterward transitions into Regime-1 dynamics and converges. The practical stakes are the convergence time, which we quantify next and which differs between the two regimes by orders of magnitude.

**Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let  $\mathbf{r}^*$  be the optimal reasoning pattern, if the overall accuracy of the initialized model  $\pi_{\theta_{\text{ref}}}$ , defined as  $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$ , satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any  $\epsilon > 0$ , there exists  $T_1 = \mathcal{O}(\frac{1}{\epsilon})$  such that for  $t > T_1$ , we have  $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$ .

Theorem 5.3 covers the strong-base-model case: whenever the reference model’s overall accuracy already exceeds the success rate of every non-optimal pattern (Eq 5.7), RLVR drives selection onto  $\mathbf{r}^*$  in time  $\mathcal{O}(1/\epsilon)$ —polynomial in  $1/\epsilon$ —and the resulting overall accuracy approaches the success rate of the optimal pattern, the best attainable under the framework.

The weak-base-model case is quantitatively very different, and matches reports that RLVR can be hard to optimize [44, 36]. Here the model must first clear an entanglement stage in which the *second-best pattern*  $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$  stalls progress toward  $\mathbf{r}^*$ :

**Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let  $\mathbf{r}^*$  and  $\mathbf{r}'$  be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(\text{Regime 2}) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R} \setminus \{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left( (C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R} \setminus \{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \quad (5.9)$$

with constants  $C_1, C_2$  depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e.,  $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$  for  $t \geq T_0$ .

Theorem 5.4 treats the case where only the optimal and second-best patterns exceed the overall accuracy; we adopt it for analytical clarity and expect it to extend, with more involved analysis, to settings where *at least* two patterns beat the average.

The quantitative punchline is the comparison of time scales. Regime 2 requires  $T_0$  steps merely to raise the overall success rate above that of the second-best pattern—i.e. to reach the Regime-1 condition of Theorem 5.3. Whereas Regime 1’s convergence time  $T_1$  is polynomial in  $1/\epsilon$ , the crossover time  $T_0$  grows *super-exponentially* in  $\gamma_{\pi_{\text{ref}}}$ , the ratio of the total success rate routed through  $\mathcal{R}/\{\mathbf{r}^*\}$  to that routed through  $\mathbf{r}^*$ . A base model that assigns tiny probability to the optimal pattern yields a large  $\gamma_{\pi_{\text{ref}}}$  and hence a prohibitive  $T_0$ : the two regimes can therefore be separated by many orders of magnitude in training time, which is precisely the entanglement stage that makes RLVR appear stuck for weak base models.

We provide additional experiments and case studies to further illustrate these two regimes in Sec 6.

## 5.4 Theoretical Explanation for RLIF

Proposition 5.2 already shows that RLIF, keyed only to the confidence of the final-answer distribution, cannot distinguish the correct answer  $\mathbf{a}^*$  from incorrect ones—so at convergence it may drop below the base model, matching the late-stage decline we measure. Yet prior work reports that RLIF also *improves* performance without external rewards Agarwal et al. [4], Zhao et al. [47], matching the early rise we measure. We now show both halves of this trajectory follow from the same analysis, quantifying each.

First, we consider a well-trained LLM satisfying the following assumption:

**Assumption 5.5.** *For the base model, the correct answer  $\mathbf{a}^*$  has the highest probability across all possible answers, i.e.,  $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$ .*

Since majority voting [35] has proven to be an effective method for improving model performance, a well-trained LLMs typically assign a higher probability to the correct answer  $\mathbf{a}^*$  than to incorrect answers, Assumption 5.5 is likely to hold for modern LLMs. Under this constraint, we consider the case that  $|\mathcal{A}| = 2$ , the success rate for each reasoning path follows a uniform distribution and the reasoning pattern selection is high-entropy<sup>2</sup>. In this case, we analyze the overall accuracy dynamics at  $t = 0$ , yielding the following theorem:

**Theorem 5.6** (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with  $\beta = 0$  for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold,  $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$  for all  $\mathbf{r} \in \mathcal{R}$ ,  $|\mathcal{A}| = 2$  and success rate for each reasoning path follows a uniform distribution  $p^*(\mathbf{r}) \sim U[0, 1]$ . Then when  $|\mathcal{R}| \rightarrow +\infty$ , the following holds:*

1. The accuracy derivative at  $t = 0$  is positive with probability 1, i.e.  $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$ .
2. With probability  $p = 0.5$ , that  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ .

The two results in Theorem 5.6 quantify the two phases exactly. First, at initialization the accuracy derivative is positive with probability 1 (in the  $|\mathcal{R}| \rightarrow \infty$  limit), so RLIF’s early gain is not incidental but essentially certain—explaining the measured initial rise. Second, as training proceeds with  $\beta = 0$ , Proposition 5.2 forces deterministic convergence to the pattern maximizing  $R_{\text{RLIF}}(\mathbf{r})$ , and this maximizer coincides with the *minimum*-success pattern,  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ , with probability exactly 0.5. Thus a coin-flip fraction of runs converges to the least accurate reasoning path. The two numbers together—near-certain early improvement, 50% eventual collapse below the base model—precisely reproduce the rise-then-degrade curve we observe empirically.

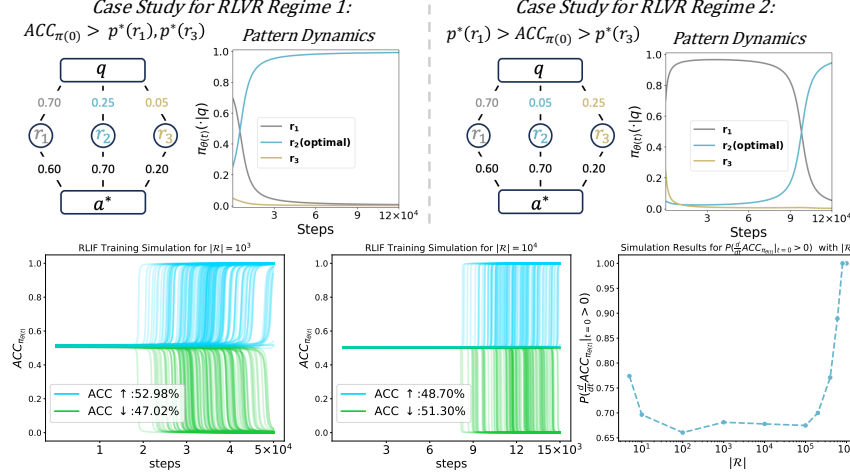


Figure 2: Direct empirical confirmation of the theory. *Upper (RLVR)*: the two predicted regimes are visibly distinct—rapid convergence to  $r^*$  (left) versus a prolonged entanglement phase before convergence (right), exactly the order-of-magnitude gap in training time predicted by Theorems 5.3 and 5.4. *Bottom (RLIF)*: the probability of converging to an improved vs. degraded state (left) and the probability of an initial accuracy gain (right), both plotted against  $|\mathcal{R}|$ ; as  $|\mathcal{R}|$  grows the improve/degrade split approaches 50/50 and the initial-gain probability approaches 1, matching the two quantitative claims of Theorem 5.6.

## 6 Case Studies and Numerical Simulations

To test the theory against measurement, we run case studies and simulations designed to reproduce each theorem’s quantitative prediction. For RLVR, Fig. 2 (upper) shows the two regimes of Theorems 5.3 and 5.4 side by side, so the predicted convergence-time gap is directly visible. For RLIF, we simulate many runs with each reasoning path’s success rate drawn uniformly and check both the initial-step behavior and the convergence outcome against Theorem 5.6, as shown in Fig. 2 (bottom).

### 6.1 Case Studies for RLVR

Section 5.3 predicts two regimes for RLVR: in Regime 1 (Eq 5.7) the selection probability  $\pi_\theta(r^*|q)$  increases monotonically to convergence, whereas in Regime 2 (Eq 5.8) an entanglement phase forces a substantially longer run before the same convergence occurs. The two panels put the predicted contrast side by side:

- When the reference model satisfies Eq 5.7 (overall accuracy above every non-optimal pattern’s success rate), Fig. 2 (upper left) shows the model locking onto the optimal pattern rapidly, as Theorem 5.3 predicts.
- When instead a suboptimal pattern  $r'$  outscores the initial overall accuracy (Theorem 5.4), the model must first traverse the entanglement phase for  $T_0$  steps. Fig. 2 (upper right) uses a large  $\gamma_{\pi_{\text{ref}}}$  (Eq 5.9):  $\gamma_{\pi_{\text{ref}}} = 6$ , since  $(0.25 + 0.05)/0.05 = 6$ . Because  $T_0$  scales like  $\gamma_{\pi_{\text{ref}}}^{\pi_{\text{ref}}}$ , this modest ratio already blows the transition time up by orders of magnitude relative to the left panel, and the measured pattern dynamics reproduce exactly this drawn-out entanglement stage.

### 6.2 Numerical Simulations For RLIF

Under Assumption 5.5 and the base-model distribution of Section 5.4, Theorem 5.6 makes two numerical predictions for RLIF in the  $|\mathcal{R}| \rightarrow \infty$  limit: a positive accuracy derivative at  $t = 0$  with probability 1, and a 50% probability of converging to the least accurate reasoning path. The simulations confirm both. In Fig. 2 (bottom left), sweeping  $|\mathcal{R}|$ , the probability of converging to an improved state ( $ACC \uparrow$ ) converges toward the probability of converging to a degraded state ( $ACC \downarrow$ )—the predicted 50/50 split. In Fig. 2 (bottom right), the probability of an initial-step

<sup>2</sup>Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case:  $\pi_{\theta_{\text{base}}}(r|q) = 1/|\mathcal{R}|$  for all  $r \in \mathcal{R}$ .

improvement rises toward 1 as  $|\mathcal{R}|$  grows, matching the first prediction. The measured curves track both theoretical values as  $|\mathcal{R}|$  increases.

## 7 Conclusions and Limitations

This work quantifies and explains reinforcement learning dynamics in LLMs by pairing direct measurement with theory. Empirically, a three-resolution comparison isolates a clean signal: RL alters the rank of only about 10% or fewer of tokens and leaves per-pattern success rates nearly flat, yet this sparse intervention reshapes pattern selection enough to move accuracy—sharply separating RLVR’s steady, monotone gains from RLIF’s rise-then-degrade trajectory. Theoretically, we reproduce these numbers for both rewards: RLVR’s convergence time splits into an  $\mathcal{O}(1/\epsilon)$  regime and a super-exponential entanglement regime, while RLIF improves at initialization with probability 1 yet converges below the base model with probability 0.5. Case studies, simulations, and experiments spanning 3B-to-32B models and multiple benchmarks all reproduce these quantitative patterns. Several limitations remain and warrant further study: the interpretability of LLM-identified reasoning patterns needs additional validation, the framework could be extended to more complex real-world reasoning, and the analysis could be generalized beyond the specific base-model assumptions used here.

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## A Additional Experiments

### A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Beyond the Number Theory tasks analyzed in Section 4.2, we repeat the same pattern-dynamics measurement on Geometry tasks (Figure 3). The contrast reproduces cleanly: under RLVR selection concentrates steadily on the highest-accuracy patterns, while under RLIF it never stabilizes—the same RLVR-vs-RLIF signature reported in the main paper, now confirmed on a second task.

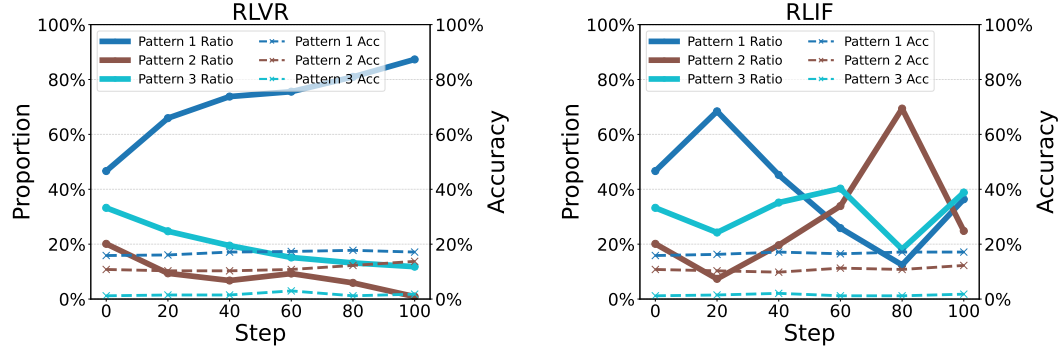


Figure 3: Pattern-distribution dynamics and per-pattern accuracy on Geometry tasks, RLVR vs. RLIF. RLVR steadily concentrates selection on the single highest-accuracy pattern (Pattern 1); RLIF never stabilizes—the same contrast seen on Number Theory in the main paper.

### A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To test whether the effect holds beyond the 3B model, we run a matched comparison at 7B (Figure 4), pairing the base model Qwen-2.5-7B-Instruct [39] against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. We broaden coverage to six settings—number theory, geometry, algebra, calculus, counting and probability from MATH [16], plus harder problems from AMC23 [1]—so the trend can be checked across domains rather than on a single task. Every setting reproduces the same direction: after RLVR the highest-accuracy patterns gain frequency and the less effective patterns lose it. Task-by-task breakdowns appear in Appendix B.3.

Across all six tasks the highest-success patterns consistently become more prevalent after RLVR, reinforcing Sections 4 and 5. The Algebra and AMC23-19 panels of Figure 4 are especially telling: per-pattern success rates stay essentially unchanged between the two models, so once again only the selection factor moves. This directly supports Assumption 5.1, on which our theoretical analysis rests.

### A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To probe the largest scale, we compare QwQ-32B [32]—built on Qwen-2.5-32B-Instruct [39] and heavily RLVR-enhanced—against its base model on four tasks: number theory and geometry from MATH [16], one task from AIME 2024 [7], and one from OlympiadBench [15].

The pattern-selection shift persists at 32B: QwQ-32B’s reasoning patterns concentrate on the patterns that are most accurate for Qwen-2.5-32B, exactly the direction the theory predicts. Here we report only the selection shift, not per-pattern accuracy: because these models are accessed through an API, we cannot extract the CoT and have the base model continue it, which is what would isolate per-pattern accuracy (rather than whole-model accuracy). See “Accuracy Analysis” in Appendix B.2 for the procedure.

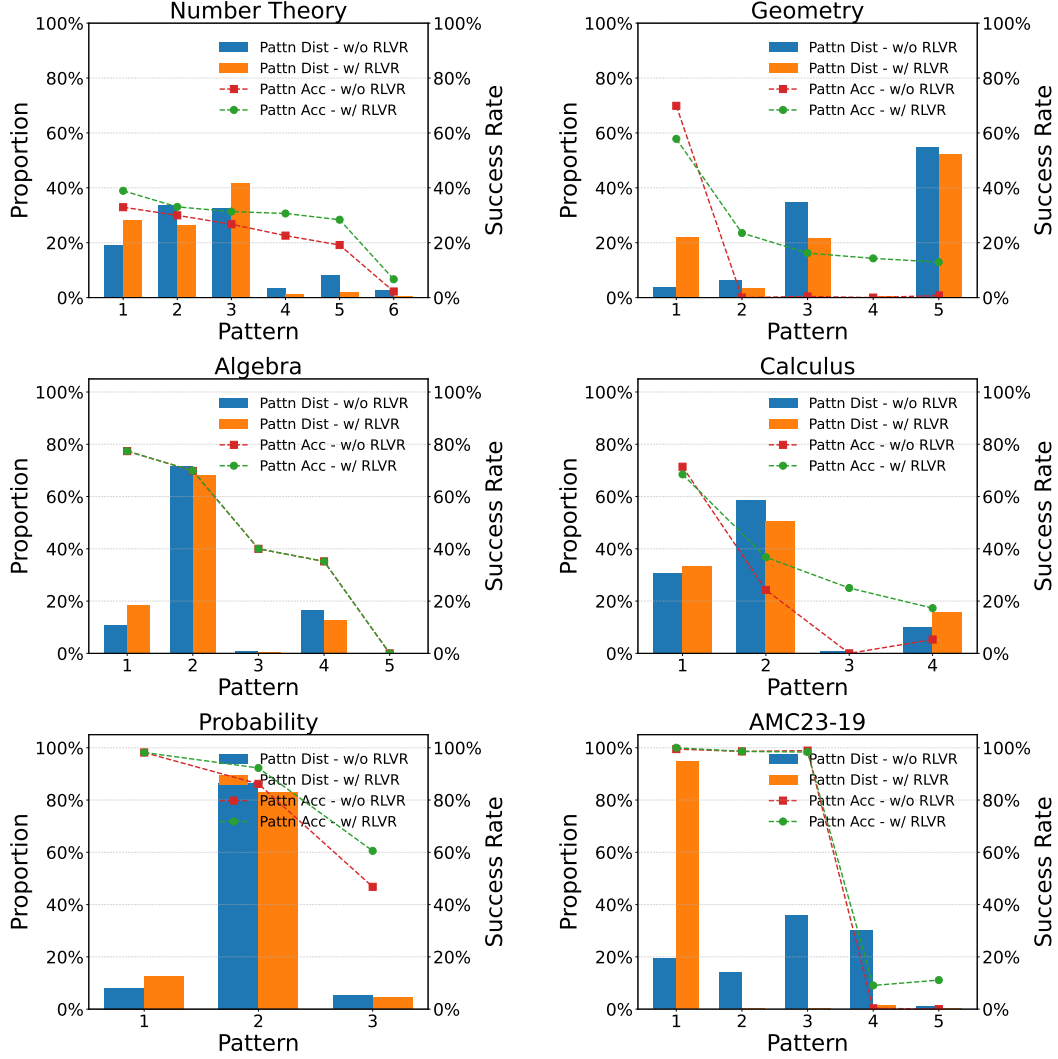
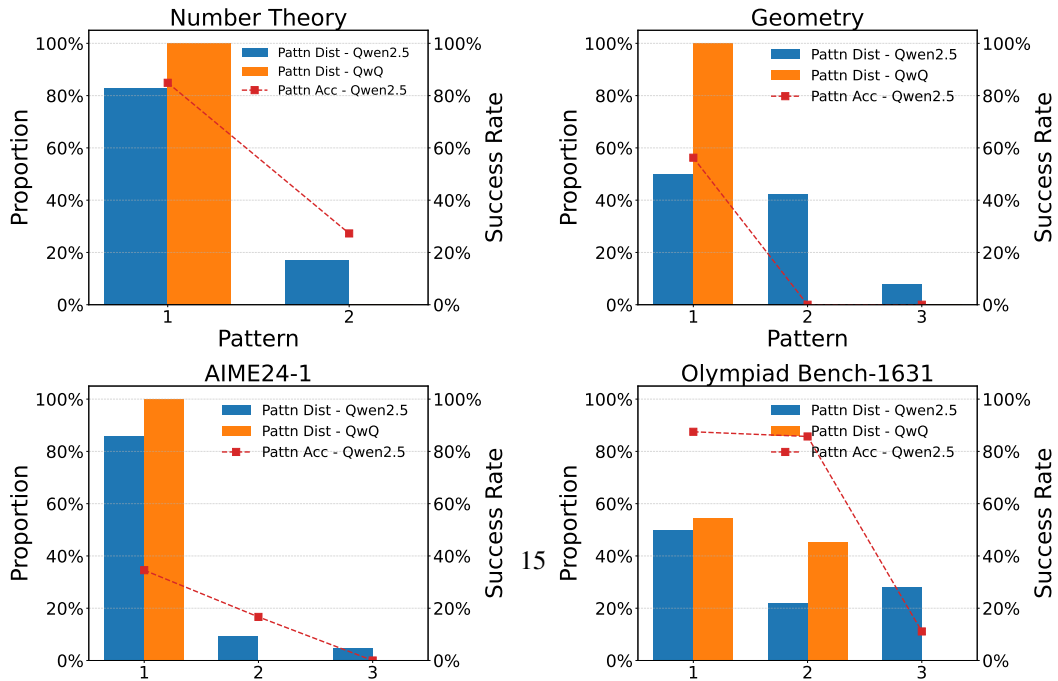


Figure 4: Reasoning-pattern distribution and per-pattern success rate, with vs. without RLVR enhancement, across six additional tasks. In every panel the mass shifts toward the higher-accuracy patterns after RLVR, matching the main-paper insights and theory. Bars (Pattern Dist) give each pattern’s share of all responses; dots (Pattern Acc) give that pattern’s success rate.



## 559 B Experimental Details

### 560 B.1 Experimental Setup

561 **Entire Training Procedures:** To make the RLVR-vs-RLIF comparison strictly controlled, we fix  
562 the base model (Qwen2.5-3B) and training set (MATH) and run both rewards under the verl [31]  
563 framework with parameter settings identical to verl’s MATH example; the reward calculation is the  
564 *only* thing changed between RLVR and RLIF. We repeat each method under three random seeds,  
565 yielding three training curves per method so the reported trajectories reflect a consistent trend rather  
566 than a single run.

567 **Token-level Analysis:** This experiment compares Qwen2.5-3B against the RL-enhanced models  
568 above, on 400 randomly sampled test questions each from GSM8K and MATH and the full AIME24  
569 set. For each question we sample the base model’s answer, concatenate it back onto the question,  
570 and feed it to the RL-enhanced model; using the OpenAI-library interface we read off each token’s  
571 probability and its rank among all tokens at that position. The comparison metric is simply the  
572 fraction of tokens whose rank differs between the base and RL-enhanced model.

573 **Reasoning-pattern Level Analysis:** We run two complementary comparisons. The first tracks  
574 the pattern distribution *during* training on our own Qwen2.5-3B and its RL-enhanced versions,  
575 checkpointed every 20 steps, so the trajectory is resolved over the whole run. The second tests  
576 generality across model scales by comparing open-source RL-enhanced models—Qwen-2.5-7B-  
577 SimpleRL-Zoo and QwQ-32B—against their respective base models; here only public checkpoints  
578 are available, so we compare the pattern distribution before vs. after RL rather than its step-by-step  
579 dynamics.

### 580 B.2 Reasoning Patterns Analysis Pipeline

581 **Model Inference:** We randomly sample questions spanning task types (Geometry, Algebra, etc.)  
582 and difficulty levels (Level 4, Level 5) from datasets including MATH. To keep the distribution  
583 estimates statistically stable, we draw 1024 answers per question for the during-training dynamics  
584 analysis, and between 64 and 1024 answers per question (by task) for the before-vs-after comparison.  
585 These responses feed the pattern extraction and classification steps.

586 **Reasoning Patterns Extraction:** We sample a subset of responses of the base model and use  
587 GPT-4o [19] API to summarize patterns categories from these samples. Full prompt we used for  
588 reasoning patterns extraction is given in Appendix D.1. To ensure the accuracy of patterns extraction,  
589 we set the temperature of GPT-4o’s API to 0.

590 **Responses Classification:** We again employ GPT-4o to classify all responses according to the  
591 identified pattern categories, also setting the temperature to 0. Full prompt we used for responses  
592 classification is given in Appendix D.2.

593 **Accuracy Analysis:** To isolate the effect of the pattern distribution on accuracy from any change  
594 in the reasoning process itself, we hold the reasoning process fixed and vary only which model  
595 completes it. Concretely, we take a base-model response, remove the sentence containing the final  
596 answer, append the preceding sentence to the question, and let the RL-enhanced model continue from  
597 that point. The accuracy of the RL-enhanced model’s continuation is our estimate of that pattern’s  
598 success rate. Because the reasoning prefix is identical across models, any accuracy difference is  
599 attributable to the pattern, not to a rewritten reasoning process—which is exactly what lets us report  
600 per-pattern success rates as a clean, comparable metric.

### 601 B.3 Details for Each Task

602 Below we detail the experimental procedures for each task, including the task descriptions and  
603 examples of reasoning patterns (Tabel 2). Example task prompts are provided in Appendix D.3.

604 **Number Theory:** This task presents models with problems involving coin distribution across  
605 multiple bags. Initially, bags contain equal numbers of coins. After receiving additional coins

| Task                     | Reasoning Patterns Examples                                     | Common Elements / Key Words                                                         |
|--------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Number Theory            | Modular Congruence with Coefficient Simplification              | Uses modular arithmetic to express divisibility conditions.                         |
|                          | Inequality-Driven Search for Minimal Solution                   | Often involves substituting back to compute the total coins.                        |
| Geometry                 | Systematic Inequality Application                               | Explicitly lists and solves each of the three inequalities.                         |
|                          | Verification of Scalene Condition                               | Adding an extra layer of validation to verifies the third side length.              |
| Algebra                  | Iterative Floor Division with Leftover Tracking.                | Uses floor division to compute new cans per step.                                   |
|                          | Recursive Recycling with Aggregated Leftovers.                  | Combines leftovers with newly produced cans before recycling.                       |
| Calculus                 | Direct Simplification and Principal Value Matching              | Assumes inputs fall within the principal range.                                     |
|                          | Interval Analysis with Case Splitting                           | Splits the whole domain into different intervals.                                   |
| Counting and Probability | Direct Probability Setup and Quadratic Solution                 | Explicitly calculates combinations for total and favorable outcomes.                |
|                          | Early Simplification and Cross-Multiplication                   | Early cross-multiplication to eliminate denominators.                               |
| AMC23-Q19                | Prime Factorization and Simplification                          | Counting Digits in Large Numbers.                                                   |
|                          | Rewriting $(8^5)$ as $(2^{15})$ $(15^5)$ as $(3^5 \cdot 5^5)$ . | Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$ . |
| AIME24-Q1                | Direct Equation Setup and Elimination                           | Straightforward and relies on algebraic manipulation.                               |
|                          | Alternative Equation Formulation and Solving                    | Expressing variables in terms of others early on.                                   |
| OlympiadBench-Q1631      | Direct Calculation and Empirical Testing                        | Relying on direct computation and empirical verification.                           |
|                          | Factorization-Based Reasoning                                   | Using algebraic factorization to argue certain terms $y_n$ .                        |

Table 2: Representative reasoning patterns per task, with their common elements and key words—the categories over which the distribution and per-pattern success-rate comparisons are computed.

and redistributing them equally, the total must exceed a specified value while maintaining equal distribution. Models must determine the minimum initial coin count per bag. We derived this task from a level 5 MATH problem (id: test/number\_theory/1055) [16], generating a 32-variant evaluation set by varying parameters such as bag count and coin totals. A one-shot prompt with a simple, unrelated example fixes the answer format ("boxed") without steering the reasoning approach. The comparison is run at matched sampling: 20 questions with 4 responses each seed the pattern extraction, and 64 responses per question are then classified into the extracted patterns for both the pre-RL and post-RL models, so the before/after distribution shift is measured on an equal footing.

**Geometry:** The geometry task gives the model two side lengths of a scalene triangle and asks how many integer values the third side can take; it is built from a level 4 MATH problem (id: test/geometry/1046). We follow the Number Theory pipeline exactly, so the two tasks are directly comparable: a 32-question synthetic set, 20 questions  $\times$  4 base-model responses for pattern extraction, and 64 responses per question over the full set for the with-vs-without-RLVR distribution comparison.

**Algebra:** The algebra task we use is a can recycling problem. We first have a certain number of old cans. It is pre-defined that  $n$  old cans can be recycled into a new can. The question is how many cans can be produced in the end. The difficulty of the problem is that in each step of the iterative calculation, there may be extra cans that cannot be divided evenly. These cans may eventually be combined together for further recycling. The template for this task comes from a level 4 algebra problem in MATH (id: test/algebra/2768). For all the following supplementary tasks starting from

this task, we adopt the same settings as the previous tasks, including the data set size, number of samples, etc.

**Calculus:** Our calculus task is simple and straightforward. We will present a trigonometric equation in a single variable, the domain of that variable, and ask the model to determine the number of solutions to the equation. Although this problem is simple, it can demonstrate the model’s basic ability in calculus problems. Its prototype is a level 4 pre-calculus problem in MATH (id: test/precalculus/1140).

**Counting and Probability:** We also study the performance of the model on the counting and probability task. There are white balls and black balls. We will randomly sample two balls from these balls and give the probability that one of the two balls drawn is black and the other is white. We also provide the number of balls of a certain color and hope that the model can calculate the minimum number of balls of another color. This is a probability theory task, which comes from a level 4 counting and probability problem in MATH (id: test/counting\_and\_probability/79).

**AMC23 Question:** Here we choose question 19 in AMC23 [1], for reasoning pattern extraction, we randomly sample 48 responses, and then sample 1024 response for reasoning pattern analysis.

#### B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

This comparison covers four tasks. Because the models are API-only, we select one question per task, draw 64 responses from each of the two models, and extract and classify reasoning patterns directly from those responses—holding the sample size matched across the two models so the distributions are directly comparable. The questions are: MATH number theory 1055 and geometry 1046, AIME24 problem 1, and OlympiadBench 1631. Representative patterns are shown in Table 2.

### C Deferred Proofs

In this appendix, we provide proofs for our main theoretical results: Proposition 5.2, Theorems 5.3, Theorems 5.4 and Theorems 5.6.

#### C.1 Proof of Proposition 5.2

*Proof of Proposition 5.2.* In this proof, we utilize the proof techniques in [26], recall that the optimization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample  $\mathbf{r}_i \in \mathcal{R}$  based on  $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$  and then provide the final answer  $\mathbf{a} \in \mathcal{A}$  by  $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$ , the reward  $\phi_{\text{RL}}(\theta)$  can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{q})} \left[ \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

Let  $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$ , then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[ R(\mathbf{r}) - \beta \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[ R(\mathbf{r}) - \beta \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[ \ln \left( \frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left( \frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

Where the third equation is by Assumption 5.1 and  $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} p^*(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$  is the partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

is a valid probability distribution such that  $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$ . Since  $Z$  is not a function of  $\mathbf{r}$ .

Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

specifically, for RLVR,  $R_{\text{RLVR}}(\mathbf{r}) = p^*$ , for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) - \underbrace{\left( \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \log |\mathcal{A}| \right)}_c, \end{aligned}$$

where the second term is the same for all  $\mathbf{r} \in |\mathcal{R}|$ , for RLIF, the optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

So we can write  $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r}))$ . This concludes the proof of the theorem.  $\square$

## C.2 Dynamics of Tabular Policy

Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the gradient dynamics of  $\theta$  can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\theta(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

Where  $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$ ,  $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$  is the column of  $\theta$  corresponding to  $\mathbf{q}$ , and  $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$  refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[ r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[ r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) (-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

where the overall accuracy is defined as  $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$ . For  $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$ , we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

### C.3 Proof of Theorem 5.3

To proof Theorem 5.3, here we first prove that  $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$  for  $t \geq 0$  (Eq C.4), then we establish a lower bound for  $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$  (Eq C.5), finally we derive the final bound for  $t$  stated in Theorem 5.3.

*Proof of Theorem 5.3.* Setting  $\beta = 0$ , Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

For  $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$ , we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

and for the accuracy  $\text{ACC}_{\theta(t)}$ :

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

As  $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$ , so  $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$  holds for  $t \geq 0$ , for the optimal reasoning pattern  $\mathbf{r}^*$ , let  $\mathbf{r}'$  the suboptimal reasoning pattern with the second highest success rate, defined as  $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$ , we have:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \quad (\text{C.5}) \end{aligned}$$

683 As  $\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$ ,  $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$  holds for  $t \geq 0$ , define  $C =$   
 684  $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$ :

$$\frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

685 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

686 For any  $\epsilon > 0$ , there exists  $T_1 = \frac{1}{C} \left( \frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$ , such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

687 This concludes the proof of the theorem.  $\square$

#### 688 C.4 Proof of Theorem 5.4

689 To proof Theorem 5.4, we first establish an upper bound for  $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$  (Eq C.6), then we analyze  
 690 the ratio  $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$  (Eq C.7) and prove that for  $t > T_0$ ,  $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$ .

691 *Proof of Theorem 5.4.* Let  $\mathbf{r}^*$  and  $\mathbf{r}'$  be the optimal and second optimal reasoning patterns. For  
 692  $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ :

$$\begin{aligned} \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\ &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2. \end{aligned}$$

693 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

694 Next, define  $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ , then for all  $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$ , we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

695 The last inequality is based on the condition for case 2 and  $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ , so  $p^*(\mathbf{r}_i) <$   
 696  $\text{ACC}_{\theta(t)}$  holds for all  $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$ ,  $t \geq 0$ . Then as  $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$ , we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

697 While  $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$ , define  $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$  we have:

$$\begin{aligned} \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

698 Then we have:

$$\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

699 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

700 Where  $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$ . Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

701 Let  $C_2 = 1/\Delta$ ,  $C_1 = p^*(\mathbf{r}')/\Delta$ , define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left( (C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

702 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left( 1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left( \frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left( \left( \frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

703 Finally, for  $t > T_0$  we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[ (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

704 This concludes the proof of the theorem.  $\square$

705 **C.5 Proof of Theorem 5.6**

706 *Proof of Theorem 5.6.* First, as we consider the case  $|\mathcal{A}| = 2$  and  $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ , Assumption  
 707 5.5 holds means  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$ . The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|q) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

708 where  $\text{Con}_{\theta(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i|q) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, q))$ . Here we assume  $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$   
 709 where  $\mathbf{a}'$  is the incorrect answer, and  $\pi_{\theta(0)}(\mathbf{r}_i|q) = \frac{1}{|\mathcal{R}|}$  for all  $\mathbf{r}_i \in \mathcal{R}$ . Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left( p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_j, q)) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left( p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left( - \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a}|\mathbf{r}_i, q)) \right). \end{aligned}$$

710 As we consider the case  $|\mathcal{R}| \rightarrow +\infty$  and the success rate  $p^*(\mathbf{r})$  are i.i.d. samples from  $U[0, 1]$ .  
 711 By the Central limit theorem, for large  $|\mathcal{R}|$ , the sample mean  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$  is concentrated  
 712 around its expectation 0.5, the deviation is of order  $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$ . We define this deviation as  
 713  $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$  such that  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$ .

714 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as  $|\mathcal{R}| \rightarrow \infty$ ,  
 715 the empirical distribution of the  $p^*(\mathbf{r})$  conditioned on their mean  $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$   
 716 converges to the density:  $f(x) = 1 + 12\delta(x - 0.5)$ ,  $x \in [0, 1]$ . As  $|\mathcal{R}| \rightarrow \infty$ , we can approximate  
 717 the summation over  $\mathbf{r}$  with an integral over this limiting distribution  $f(x)$ . The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

718 Since  $\delta > 0$ , the derivative is positive. As  $|\mathcal{R}| \rightarrow \infty$ , this holds with probability 1, proving the  
 719 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size  
 $|\mathcal{A}| = 2$ , the reward for a path  $\mathbf{r}$  is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

720 The function  $g(p) = p(1 - p)$  is maximized at  $p = 0.5$ . Therefore, maximizing the reward  $R_{\text{RLIF}}(\mathbf{r})$   
 721 is equivalent to choosing the path whose success rate  $p^*(\mathbf{r})$  is furthest from 0.5.

Let  $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$  and  $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ . The path with the highest reward will be the  
 one with success rate  $p_{\min}^*$  if it is further from 0.5 than  $p_{\max}^*$  is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

722 Thus, the event  $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$  is equivalent to the event  $p_{\min}^* + p_{\max}^* < 1$ .

723 We now calculate the probability of this event. The success rates  $p^*(\mathbf{r})$  are drawn from the limiting  
 724 distribution  $f(x) = 1 + 12\delta(x - 0.5)$ . From the theory of order statistics, the scaled minimum and  
 725 maximum of  $n = |\mathcal{R}|$  samples converge in distribution to independent exponential random variables.  
 726 Let  $Y_n = n \cdot p_{\min}^*$  and  $Z_n = n \cdot (1 - p_{\max}^*)$ . As  $n \rightarrow \infty$ :

- 727 •  $Y_n$  converges to  $Y \sim \text{Exp}(\lambda_Y)$ , where the rate is the density at the lower bound:  $\lambda_Y = f(0) =$   
728  $1 - 6\delta$ .  
729 •  $Z_n$  converges to  $Z \sim \text{Exp}(\lambda_Z)$ , where the rate is the density at the upper bound:  $\lambda_Z = f(1) =$   
730  $1 + 6\delta$ .

The condition  $p_{\min}^* + p_{\max}^* < 1$  can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is  $P(Y < Z)$ . For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since  $\delta = O(1/|\mathcal{R}|^{1/2})$ , in the limit as  $|\mathcal{R}| \rightarrow \infty$ ,  $\delta \rightarrow 0$ . Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta\right) = \frac{1}{2}$$

731 This concludes the proof of the second statement.

732

□

## 733 D Full Prompts

### 734 D.1 Full Prompt for Reasoning Patterns Extraction

#### Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

735

Example processes from input  
Common elements  
Pattern 2:  
Name and description  
Defining characteristics  
Example processes from input  
Common elements  
...  
There are reasoning processes:  
{extracted\_responses}  
Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

736

## 737 D.2 Full Prompt for Responses Classification

### Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.  
{patterns\_categories}  
Here are the reasoning processes to classify:  
{extracted\_responses}  
Please output your classification in this JSON format:

738

## 739 D.3 Prompt Examples for Evaluation Tasks

### Prompt Example for Easy Countdown

<lim\_start>system  
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim\_endl>  
<lim\_start>  
user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim\_endl>  
<lim\_start>assistant  
Let me solve this step by step.  
<think>

740

### Prompt Example for Long Arithmetic

<lim\_start>system  
You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim\_endl>  
<lim\_start>user  
Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim\_endl>  
<lim\_start>assistant  
Let me solve this step by step.  
<think>

741

#### Prompt Example for Number Theory

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

742

#### Prompt Example for Geometry

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
Two sides of scalene  $\triangle ABC$  measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

743

#### Prompt Example for Algebra

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

744

#### Prompt Example for Calculus

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
For how many values of  $x$  in  $[0, 2\pi]$  is  $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$ ?  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

745

#### Prompt Example for Counting and Probability

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
7 white balls and  $k$  black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is  $\frac{35}{66}$ . Find the smallest possible value of  $k$ .  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

746

Prompt Example for AMC23-Q19

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
How many digits are in the base-ten representation of ' $8^5 \cdot 5^{10} \cdot 15^5$ '?  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

747

Prompt Example for AIME24-Q1

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
Every morning Aya goes for a  $9$ -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of  $s$  kilometers per hour, the walk takes her  $4$  hours, including  $t$  minutes spent in the coffee shop. When she walks  $s+2$  kilometers per hour, the walk takes her  $2$  hours and  $24$  minutes, including  $t$  minutes spent in the coffee shop. Suppose Aya walks at  $s+\frac{1}{2}$  kilometers per hour. Find the number of minutes the walk takes her, including the  $t$  minutes spent in the coffee shop.  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

748

Prompt Example for OlympiadBench-Q1631

<lim\_start>system  
You are a helpful assistant.<lim\_end>  
<lim\_start>user  
For a positive integer  $a$ , define a sequence of integers  $x_1, x_2, \dots$  by letting  $x_1 = a$  and  $x_{n+1} = 2x_n + 1$  for  $n \geq 1$ . Let  $y_n = 2^{x_n} - 1$ . Determine the largest possible  $k$  such that, for some positive integer  $a$ , the numbers  $y_1, \dots, y_k$  are all prime.  
Please reason step by step, and put your final answer within  $\boxed{\phantom{000}}$ .<lim\_end>  
<lim\_start>assistant

749